

Reviewer Pack — Minimal Topological Entropy Bound on Communication Complexit...

Entry 32c4b005b684 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Topological Entropy Bound on Communication Complexity Rank Variance

Entry ID: 32c4b005b684 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-07 10:35:17 UTC

1. Conjecture

Field A (mathematical branch): Topological Dynamics **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

For any given CNF φ with n variables, the communication complexity rank variance ($\text{CRV}(\varphi)$) of φ is upper-bounded by the minimal topological entropy ($H_{\min}(\varphi)$) of φ , such that $\text{CRV}(\varphi) = O(H_{\min}(\varphi)^2)$.

Rationale (proposer's reasoning):

Topological dynamics can provide a measure of information flow in a system. If the communication complexity rank variance captures the complexity of distributing information among parties, then a lower entropy could imply a more efficient distribution, leading to a bound on the communication complexity rank variance.

Taxonomy category: Topological_Dynamics (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 2f6923e903c9eaeef

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a given CNF φ with n variables, if the communication complexity rank variance ($\text{CRV}(\varphi)$) is within the range $[O(H_{\min}(\varphi)^2), 1.5 * O(H_{\min}(\varphi)^2)]$ for all seeds and $H_{\min}(\varphi) \leq \log_2(n^3)$, then the conjecture is supported.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	1.00	UNCERTAIN	SAFE
KARP_LIPTON	SAFE	1.00	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "topological dynamics" AND "communication complexity rank variance" - "minimal topological entropy" AND communication complexity matrix rank" - "upper bound on CRV" AND H_{min} in topological dynamics

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(2**n):
            clause = [random.choice([-1, 1]) * (i + 1) for i in range(n)]
            if all(clause[i] != -clause[j] for i in range(n) for j in range(i+1, n)):
                clauses.append(clause)
        return clauses

    def min_topological_entropy(cnf):
        # Placeholder implementation
        # This is a dummy function and should be replaced with actual computation
        return random.random()

    def communication_complexity_rank_variance(cnf):
        # Placeholder implementation
        # This is a dummy function and should be replaced with actual computation
        return random.random() * min_topological_entropy(cnf)

    n = 10 # Example value for n, can be adjusted within the loop
    crv = communication_complexity_rank_variance(generate_cnf(n))
    h_min = min_topological_entropy(generate_cnf(n))

    if h_min <= math.log2(n**3):
        conjecture_holds = crv <= 1.5 * h_min**2
    else:
        conjecture_holds = False
```

```

return {
    "metric_name": "communication_complexity_rank_variance",
    "metric_value": crv,
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": conjecture_holds,
    "counterexample": "" if conjecture_holds else "mapping_undefined"
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_crv = sum(r["metric_value"] for r in results) / len(results)
    std_crv = math.sqrt(sum((r["metric_value"] - mean_crv)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_crv} std={std_crv} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

: ''}
TRIAL: {'metric_name': 'communication_complexity_rank_variance', 'metric_value': 0.0197134439722439}
TRIAL: {'metric_name': 'communication_complexity_rank_variance', 'metric_value': 0.3085260838243877}
TRIAL: {'metric_name': 'communication_complexity_rank_variance', 'metric_value': 0.0233021198692501}
TRIAL: {'metric_name': 'communication_complexity_rank_variance', 'metric_value': 0.0088520810321546}
TRIAL: {'metric_name': 'communication_complexity_rank_variance', 'metric_value': 0.0599713025823159}
TRIAL: {'metric_name': 'communication_complexity_rank_variance', 'metric_value': 0.0019164869977440}
TRIAL: {'metric_name': 'communication_complexity_rank_variance', 'metric_value': 0.2379676756531177}
TRIAL: {'metric_name': 'communication_complexity_rank_variance', 'metric_value': 0.0454248237117482}
TRIAL: {'metric_name': 'communication_complexity_rank_variance', 'metric_value': 0.7096296204085583}
TRIAL: {'metric_name': 'communication_complexity_rank_variance', 'metric_value': 0.0210717591828642}
RESULT: FALSIFIED counterexample="mapping_undefined" first_failing_seed=191

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test code uses placeholder implementations for both the minimal topological entropy and communication complexity rank variance calculations, which are not faithful to their

mathematical definitions. The conjecture’s validity cannot be confirmed with such placeholders.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code uses placeholder implementations for both the minimal topological entropy and communication complexity rank variance calculations, which are not faithful to their mathematical definitions. The critic has challenged the validity of the test results. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15798
2	preregistration	ollama_remote	glm4:latest	0	0	9617
3	novelty	ollama_remote	glm4:latest	0	0	8250
4	novelty	ollama_remote	glm4:latest	0	0	8691
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	25417
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11691
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12006
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7738
9	critic	ollama_remote	glm4:latest	0	0	20798
10	judge	ollama_remote	glm4:latest	0	0	8772

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 128778 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/32c4b005b684.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/32c4b005b684.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/32c4b005b684.tar.gz (if generated)